

# IT Applications in Business

## Blockchain and cryptocurrencies

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HR EXCELLENCE IN RESEARCH



Politechnika Wroclawska

# Lecture 1: ***Blockchain and Cryptocurrencies***

## **1. History**

## **2. Blockchain essentials**

- Distributed / decentralized systems
- Block structure
- Hash function
- Merkle Trees
- Ledger

## **3. Consensus algorithm - Proof of work**

## **4. Blockchain - why to use ?**

## ***A bit of history***

- **1998** - idea of cryptocurrency described **Wei Dai** on the *cypher punks* mailing list
- **2007** - Satoshi Nakamoto – conceptualized a new system that constituted a base for Blockchain and Bitcoin
- **Aug 2008** - Bitcoin.org established and registered

# Bitcoin Whitepaper

- <https://bitcoin.org/bitcoin.pdf> - **2008.10.31**

[PDF] **Bitcoin: A peer-to-peer electronic cash system**

S Nakamoto - Decentralized business review, 2008 - [assets.pubpub.org](https://assets.pubpub.org)

A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures ...

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## Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto  
[satoshin@gmx.com](mailto:satoshin@gmx.com)  
[www.bitcoin.org](http://www.bitcoin.org)

**Abstract.** A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest

## ***A bit of history***

- **Dec 2009** - first Bitcoin transaction occurs
- **May 2010** - first real world transaction (***Laszlo Hanyecz*** -> 10k BTC for 2 pizzas\*)

<https://bitcointalk.org/index.php?topic=137.msg1195>

\* Value of \$41 back then



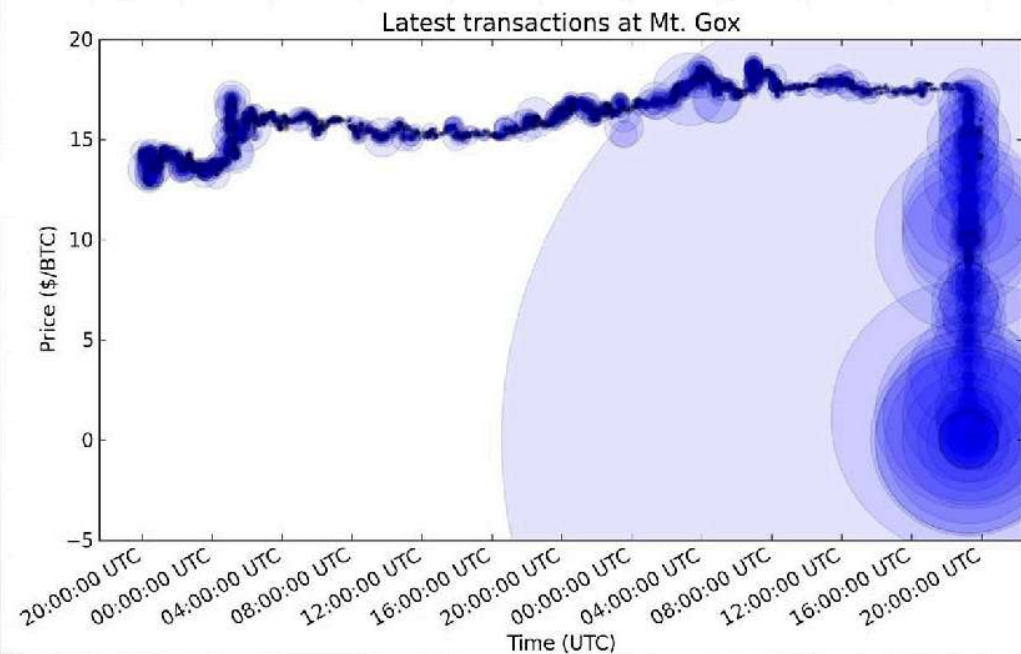
## ***A bit of history***

- **July 2010** - *Mt.Gox* a bitcoin exchange based in Tokyo, Japan
- **Feb 2011** - BTC reaches *parity* with USD
- **2011** - BTC has started gaining momentum, *Silk Road* opens



## *A bit of history*

- **2011** - security breach at Mt. Gox - nominal price of BTC fraudulently drops to 1 US cent



## *A bit of history*

- **Jun 2012** - Block 181919 is the largest block, consists of 13222 transactions
- **Jun 2012** - *coinbase*, a BTC wallet in SF
- **May 2013** - first BTC ATM in San Diego



## *A bit of history*

- **Oct 2013** - Silk Road closed by FBI
- **Aug 2013** - NY State Dept issues subpoenas to 22 BTC companies
- **Nov 2013** - BTC breaks 1k USD barrier
- **Feb 2014** - massive attacks on exchanges (Mt. Gox collapses) → BTC plummets
- **Sept 2014** - PayPal integrates with BTC



## A bit of history

- Dec 19<sup>th</sup>, 2017 – 1 BTC reaches record high value **18402,25 USD**
- Dec 2018 – (year to year) value of cryptocurrencies continues to depreciate
  - BTC -> **-82%**
  - BTC Cash -> **- 94,4%**
  - Ethereum -> **-88,84%**
  - Litecoin -> **-92.1%**

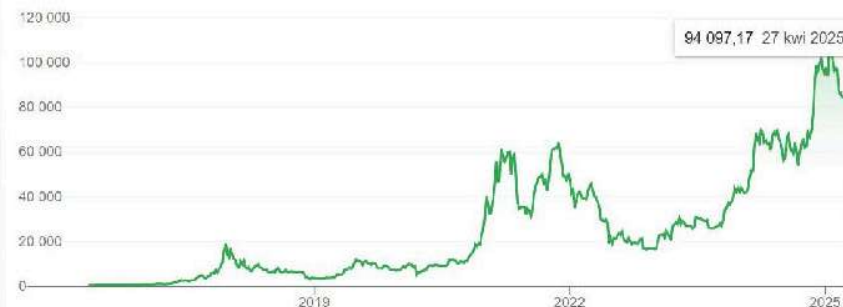


It took just five days for bitcoin to rise from \$6,000 to \$8,000

94 097,17 USD  
+93 643,18 (20 626,71%) ↑ zawsze

27 kwi, 10:31 UTC - Wyłączenie odpowiedzialności

1D 5D 1M 6M YTD 1R 5L Maks.



## A bit of history



### BIG HACKS OF MAJOR EXCHANGES

2014

MT. GOX



**\$473 MILLION**

Allegedly stolen by Russian hacker and owner of BTC-e exchange, Alexander Vinnik

2016

BITFINEX



**\$72 MILLION**

Stolen through vulnerabilities in multi-signature wallet

2018

COINCHECK



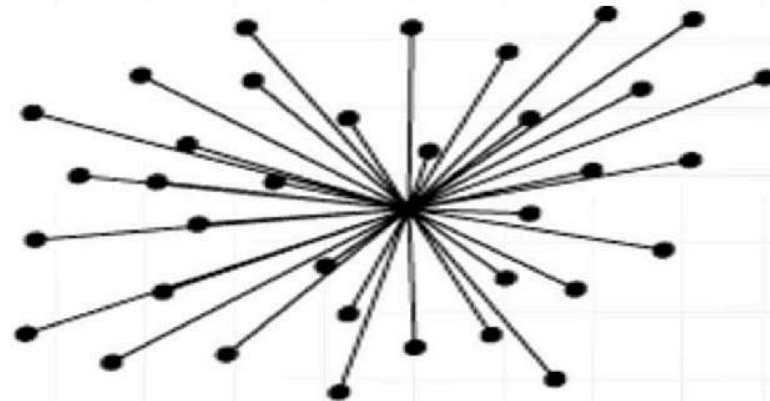
**\$530 MILLION**

Stolen due to hot wallet storage without multi-signature protection

# Centralized vs. Decentralized vs. Distributed

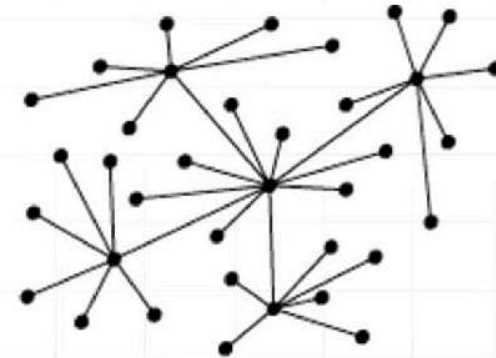
- **Centralized**

- one authority controls everything,
- if it goes offline, peers lose access to services



# Centralized vs. Decentralized vs. Distributed

- **Decentralized**
  - some nodes have control on services / process information



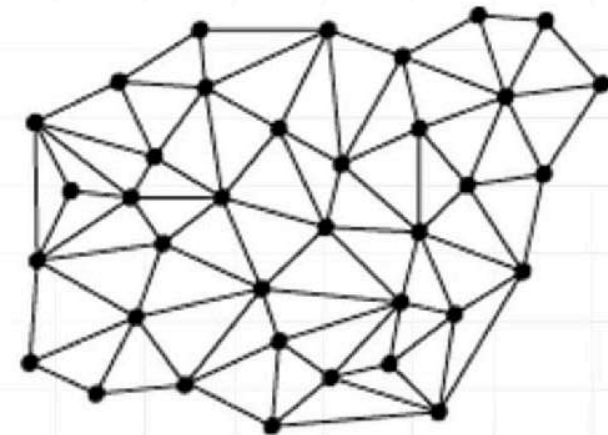
# Centralized vs. Decentralized vs. Distributed

- **Distributed**

- no dedicated node to process data

- **Advantages**

- Higher computing power
  - Cost reduction
  - Higher reliability
  - Ability to grow naturally



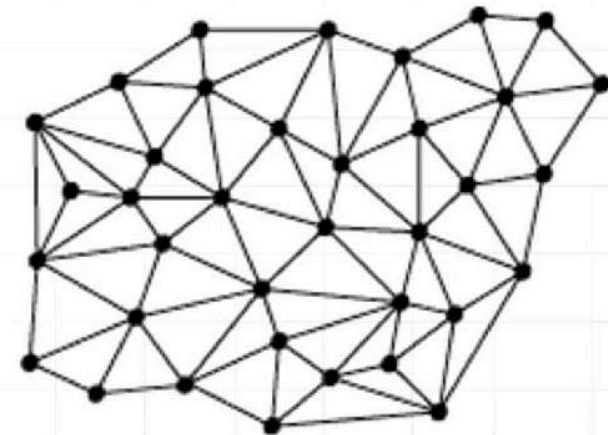
# Centralized vs. Decentralized vs. Distributed

- **Distributed**

- no dedicated node to process data

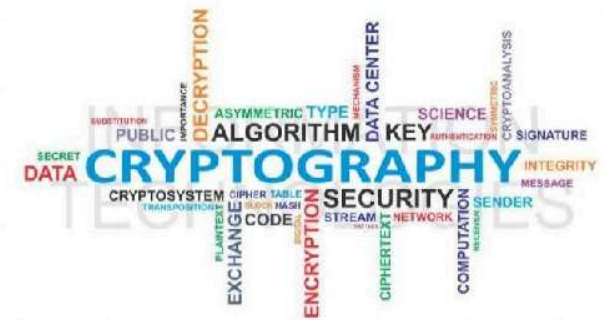
- **Disadvantages**

- Coordination overhead
  - Communication overhead
  - Dependency on networks
  - Higher program complexity
  - Security issues



# Cryptography

- foundation of cryptocurrencies
- earliest known use of cryptography - carved ciphertext on stone Egypt (ca 1900 BCE)
- Greek *kryptós*, "hidden, secret"; and *graphein*, "to write", or *-logia*, "study",
- a mathematical algorithm to secure data
- **Encryption** – “human readable\*” > encrypted form
- **Decryption** – encrypted form > “human readable”



# Cryptography

## 1. Cipher

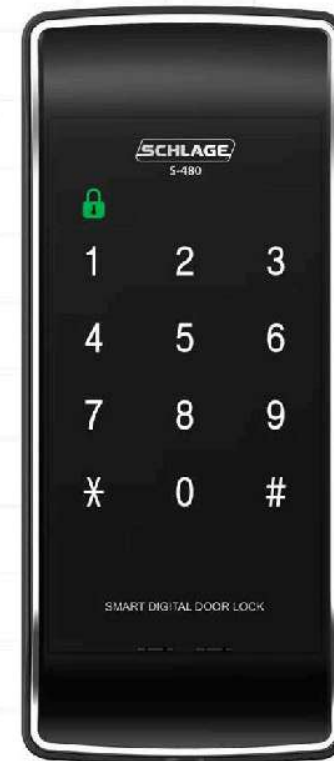
- rules to encode the information

## 2. Key

- allows to encrypt / decrypt the data

## 3. Cryptography

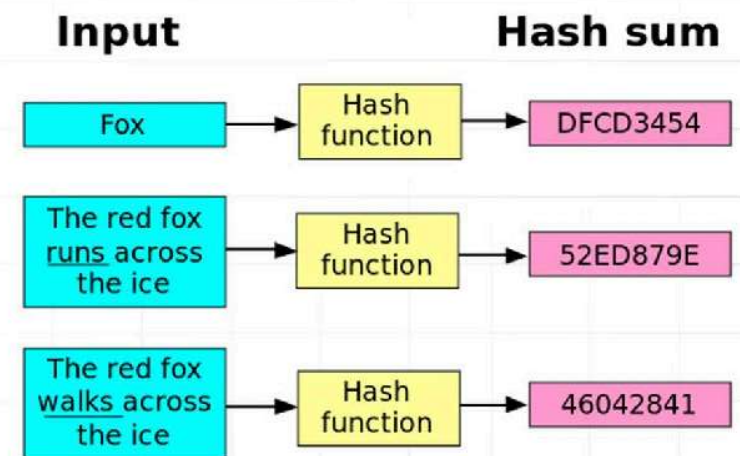
- uses (1) as a digital lock and (2) to *encrypt* and *decrypt* Data



# Hash

- Hashing algorithms (MD5\*, SHA-1, SHA-2, SHA-3)
  - need to reasonable fast to get hashed value
  - one bit change should result into entirely different hash value
  - must be unique to prevent hash collisions

\*deprecated, vulnerable, can be cracked by brute force



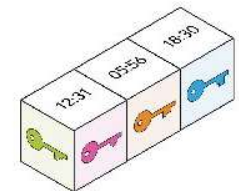
# Hash

| Algorithm and variant |                            | Output size (bits)             | Internal state size (bits) | Block size (bits) | Rounds                             | Operations                                         | Security against collision attacks (bits) | Security against length extension attacks (bits) | Performance on Skylake (median cpb) <sup>[1]</sup> |                  | First published |
|-----------------------|----------------------------|--------------------------------|----------------------------|-------------------|------------------------------------|----------------------------------------------------|-------------------------------------------|--------------------------------------------------|----------------------------------------------------|------------------|-----------------|
|                       |                            |                                |                            |                   |                                    |                                                    |                                           |                                                  | Long messages                                      | 8 bytes          |                 |
| MD5 (as reference)    |                            | 128                            | 128<br>(4 × 32)            | 512               | 4<br>(16 operations in each round) | And, Xor, Or, Rot, Add (mod 2 <sup>32</sup> )      | ≤ 18<br>(collisions found) <sup>[2]</sup> | 0                                                | 4.99                                               | 55.00            | 1992            |
| SHA-0                 |                            | 160                            | 160<br>(5 × 32)            | 512               | 80                                 | And, Xor, Or, Rot, Add (mod 2 <sup>32</sup> )      | < 34<br>(collisions found)                | 0                                                | ≈ SHA-1                                            | ≈ SHA-1          | 1993            |
| SHA-1                 |                            |                                |                            |                   |                                    |                                                    | < 63<br>(collisions found) <sup>[3]</sup> |                                                  | 3.47                                               | 52.00            | 1995            |
| SHA-2                 | SHA-224                    | 224                            | 256<br>(8 × 32)            | 512               | 64                                 | And, Xor, Or, Rot, Shr, Add (mod 2 <sup>32</sup> ) | 112                                       | 32<br>0                                          | 7.62                                               | 84.50            | 2004            |
|                       | SHA-256                    | 256                            |                            |                   |                                    |                                                    | 128                                       |                                                  | 7.63                                               | 85.25            | 2001            |
|                       | SHA-384                    | 384                            | 512<br>(8 × 64)            | 1024              | 80                                 | And, Xor, Or, Rot, Shr, Add (mod 2 <sup>64</sup> ) | 192                                       | 128                                              | 5.12                                               | 135.75           | 2001            |
|                       | SHA-512                    | 512                            |                            |                   |                                    |                                                    | 256                                       | 0 <sup>[4]</sup>                                 | 5.06                                               | 135.50           | 2001            |
|                       | SHA-512/224<br>SHA-512/256 | 224<br>256                     |                            |                   |                                    |                                                    | 112<br>128                                | 288<br>256                                       | ≈ SHA-384                                          | ≈ SHA-384        | 2012            |
| SHA-3                 | SHA3-224                   | 224                            | 1600<br>(5 × 5 × 64)       | 1152              | 24 <sup>[5]</sup>                  | And, Xor, Rot, Not                                 | 112                                       | 448                                              | 8.12                                               | 154.25           | 2015            |
|                       | SHA3-256                   | 256                            |                            | 1088              |                                    |                                                    | 128                                       | 512                                              | 8.59                                               | 155.50           |                 |
|                       | SHA3-384                   | 384                            |                            | 832               |                                    |                                                    | 192                                       | 768                                              | 11.06                                              | 164.00           |                 |
|                       | SHA3-512                   | 512                            |                            | 576               |                                    |                                                    | 256                                       | 1024                                             | 15.88                                              | 164.00           |                 |
|                       | SHAKE128<br>SHAKE256       | d (arbitrary)<br>d (arbitrary) |                            | 1344<br>1088      |                                    |                                                    | min(d/2, 128)<br>min(d/2, 256)            | 256<br>512                                       | 7.08<br>8.59                                       | 155.25<br>155.50 |                 |

# ***Blockchain - essentials***



- Blockchain is NOT Bitcoin (sic!)
- Core foundation of a number of cryptocurrencies
- A transaction system
- Some sort of decentralized database
- No one has control over transactions
- Unhackable / tamper proof



coindesk

# ***Blockchain - essentials***

- ***Ledger***

- a way of recording transactions
- (time, date, participants, transaction)

- ***Transactions***

- all traceable - unlike to traditional monetary transactions (non electronic)

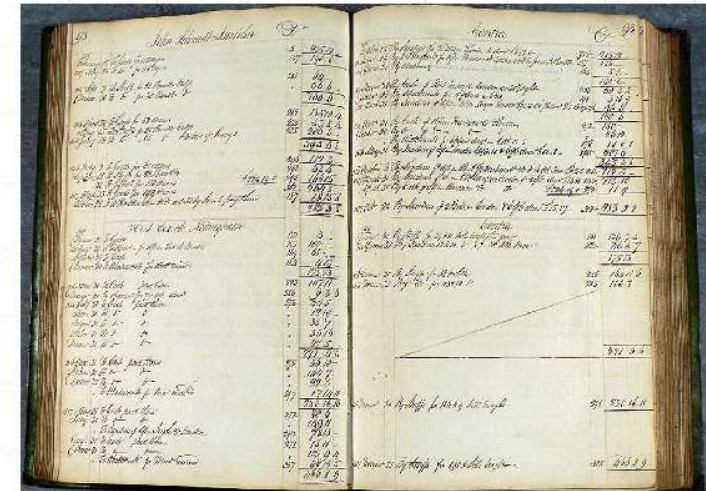
- ***Exchange***

- essentially bank-like entity (centralized)
- an intermediary



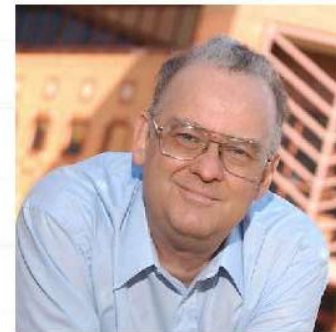
# Ledger

- In finance - a book that holds transactions
- a **public** (!), decentralized, distributed database
- store all the transactions ever occurred (and will occur)
- INTERNET connection **required** to view it
- in simplification (ID, Sender, Receiver, Timestamp)



# ***Merkle Tree***

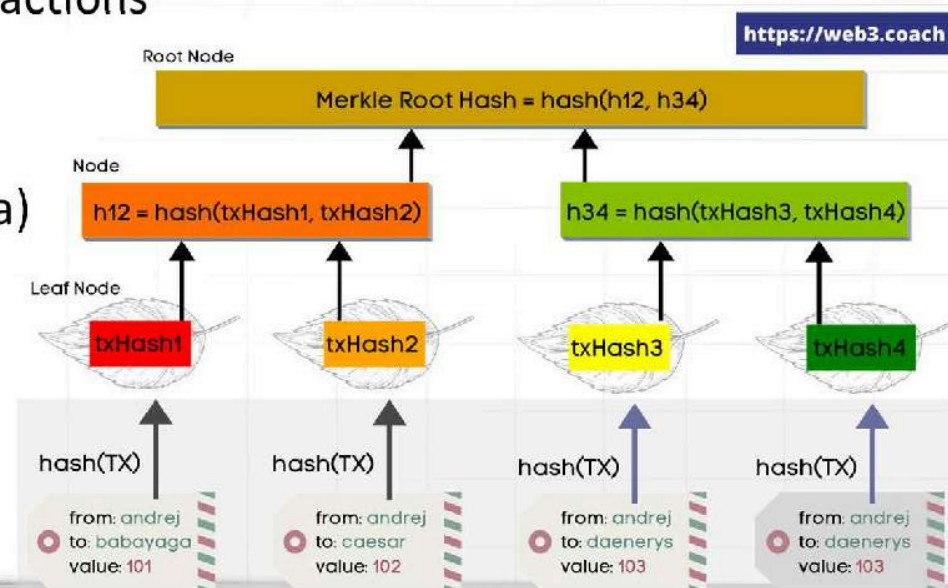
- key part of blockchain
- related to hashing algorithm(s)
- takes an input and help confirms its validity
- each transaction has its hash
- concept - *Ralph Merkle\** (patented in 1979)



# Merkle Tree

- hashes are combined into pairs -> a **combination hash** (hash of hashes)
- ROOT (Top) hash is an eventual hash of all transactions
- prevent tamper with transactions\*
- **MT** allows checking validity of transactions (data)

\* One sub of hashes altered = results in cascade effect (avalanche)



# Blockchain - essentials



- **Blockchain** is a distributed transaction ledger (register)
- Transactions are grouped in **blocks**, one by one
- System is based on **peer-to-peer**, without central nodes
- Integrity is ensured by **hashing** functions (SHA-2 / SHA-3)
- Authentication ensured by **electronic signature** algorithms
- **Transactions** are gathered in **blocks** using broadcasting

## ***Blockchain - essentials***



- Each node can participate in adding a block to the ledger
- A **reward** driven system for achieving **consensus** based on “***Proof of Work***” (***mining***) for helping secure the network
- For a job done by a node, it can be rewarded a ***transaction fee***
- Authentication ensured by ***electronic signature*** algorithms
- A **publicly** disclosed linked **ledger** of transactions stored in a blockchain

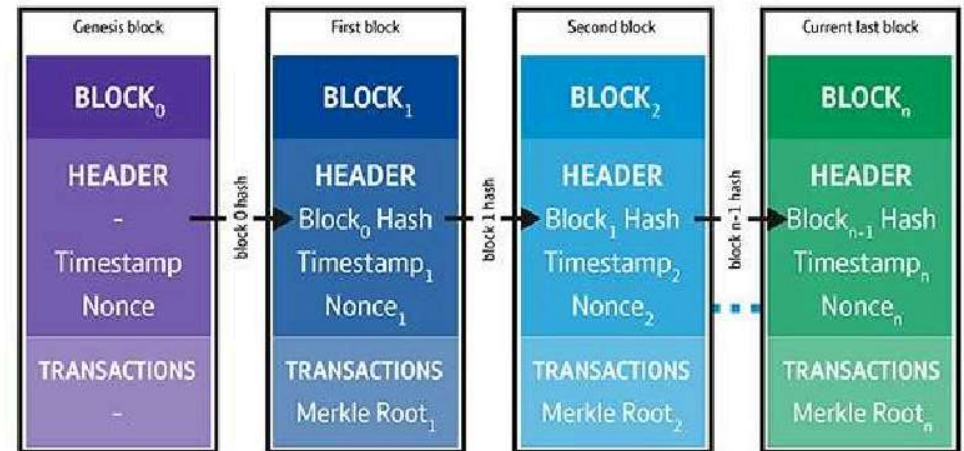
# Block

- **block** is a container for transactions, which are each signed by both respective parties
- **block hash (signature)** – a sort of lock, locking the block and preventing further changes of the transactions stored inside.
  - the *previous block hash*,
  - the hashes of the transactions contained inside,
  - the block's creation time,
  - a random string: the nonce.

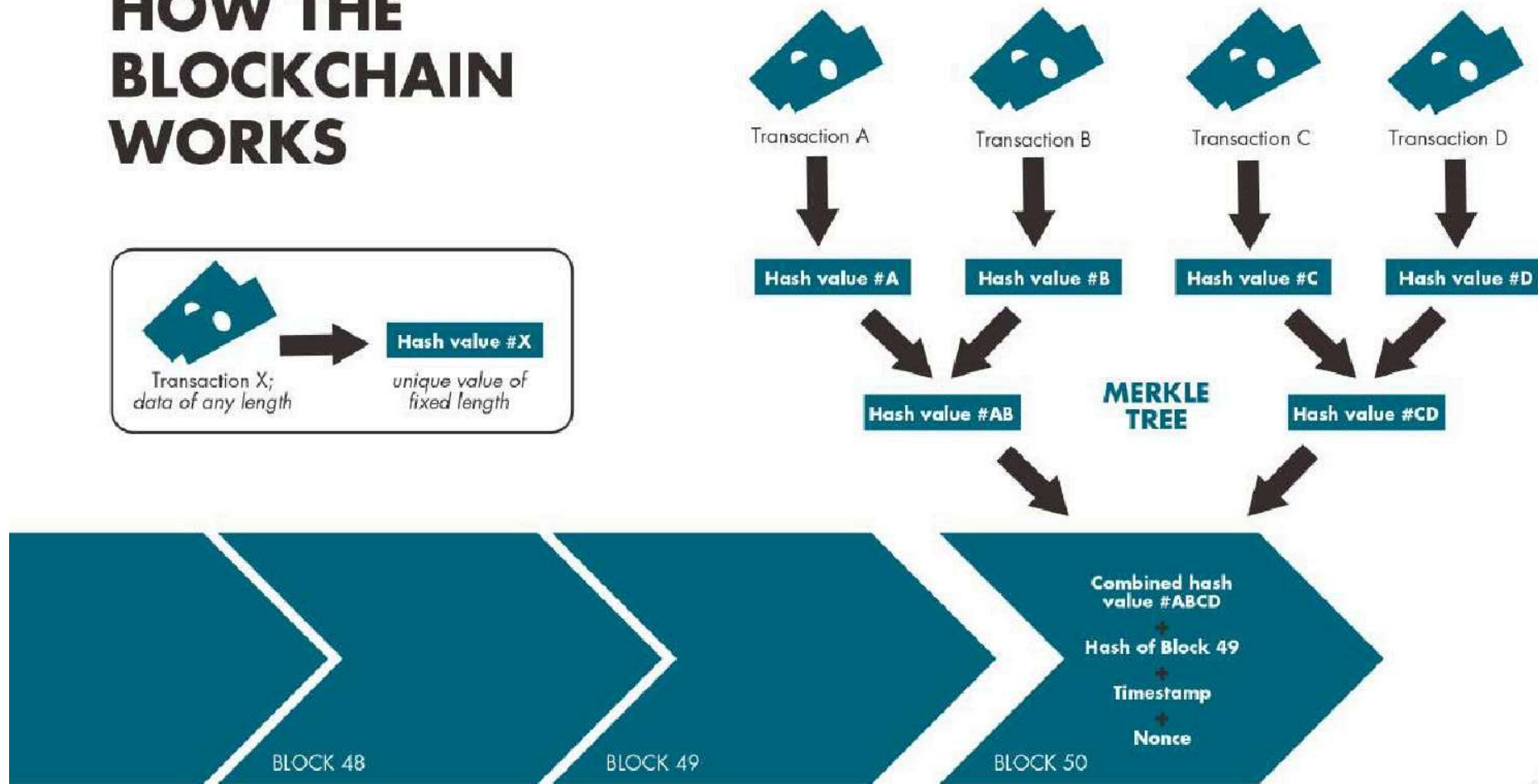
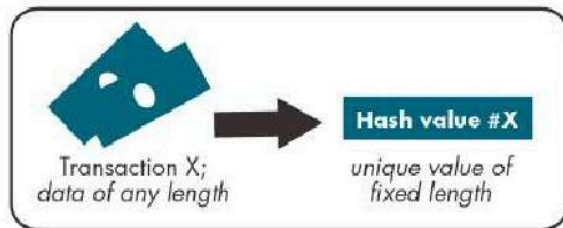


## Block header

- Version
- Previous Block Hash
- Merkle root (hash of Merkle tree)
- Timestamp (approx. creation of transaction)
- Difficulty Target (a proof of work algorithm for this block)
- Nonce (essentially is a random number)



# HOW THE BLOCKCHAIN WORKS



Reproduction of an original figure in "The Great Chain of Being Sure About Things" by the Economist

Center  
of Global  
Development

## (Full) Block chain

- chain formed by all blocks linked in sequence
- a new block is appended (mined) roughly every **10** minutes (Proof-of-Work)
- uses the **block hash** of the previous block to create its own **block hash**
- blocks increasingly harder to corrupt as more blocks are added on

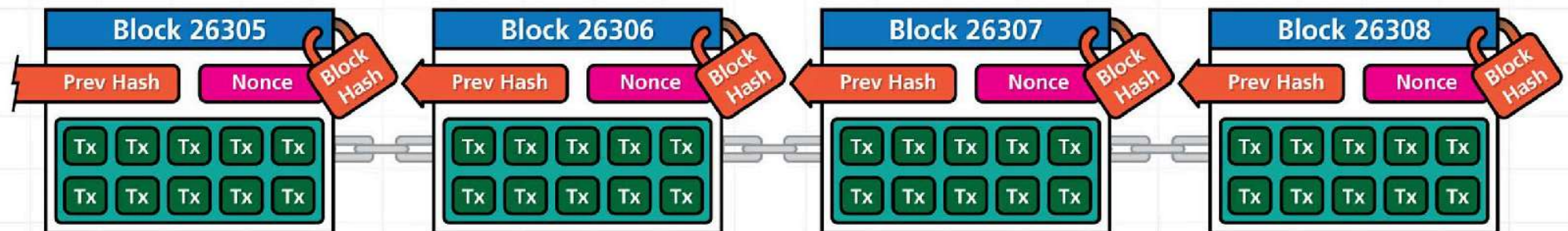


Illustration by CryptoGraphics.info

## *(Non-full node) Block chain*

- non-full node runs on a 'light' blockchain copy
  - instead of all the transactions – each block only contains the **Merkle Root**
- a way to verify the entire blockchain
  - without the body of every transaction in the block header

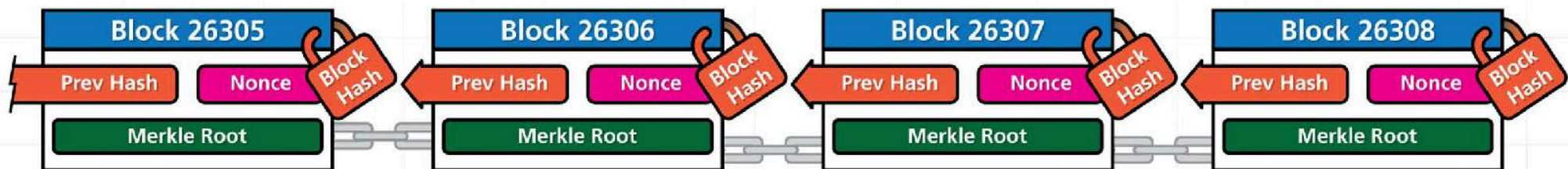
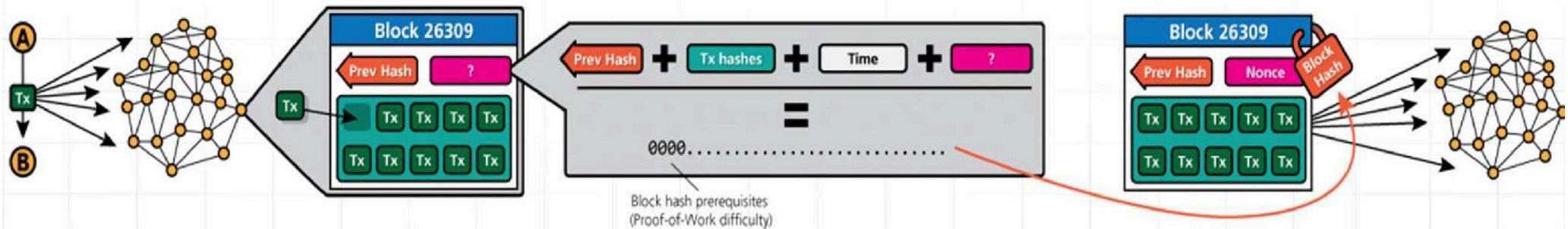


Illustration by CryptoGraphics.info

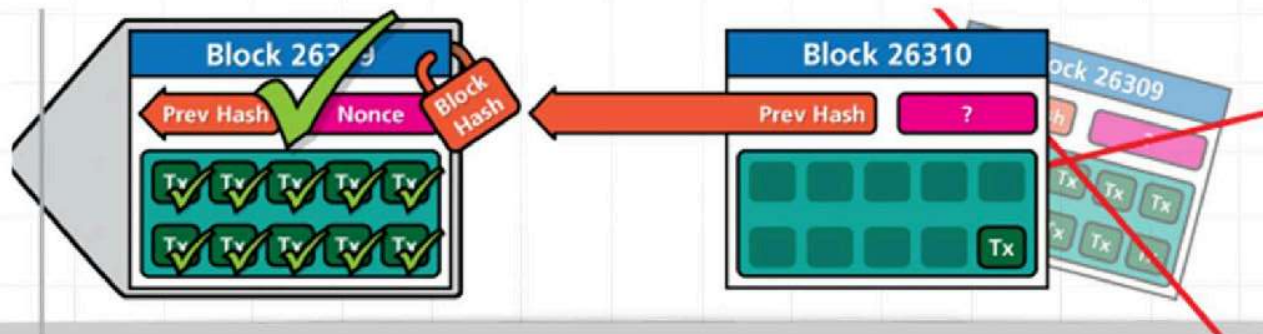
# Block chain – in a nutshell

- new transactions are sent to all nodes (broadcast)
- each node collect new transactions into a block
- once filled, each node works on finding **nonce** (PoW) for a block → calculates **block hash**
- when a node, finds correct **nonce**, it sends the signed block to ALL nodes (broadcast)



## Block chain – Consensus mechanism

- a node that found correct nonce is **rewarded**
- nodes accept the block only if **ALL** transactions are VALID
- nodes express acceptance of the block by discarding their own or (unfinished) block
- nodes start working a new block – using block hash of the accepted block as previous hash



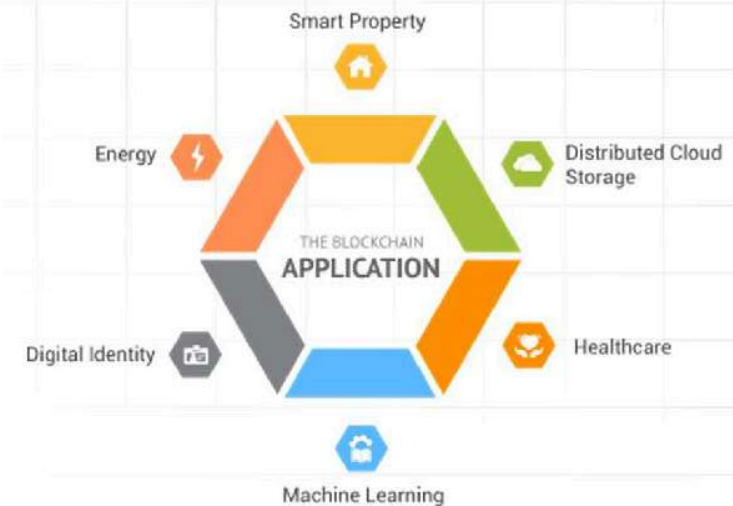
## Blockchain – why use it?

- Not controlled by neither:
  - a corporation nor government
- Secure
  - no single point of failure
- Open
  - royalty free
- Provides complete audit trail



# ***Blockchain – applications***

1. Storage for digital records
2. Exchanging digital assets (called tokens)
3. Executing smart contracts
  - Ground rules – Terms & conditions recorded in a code
  - Distributed network executes contract & monitors compliance
  - Outcomes are automatically validated without third party



Tech Trends 2017, The Kenetic Enterprise, "Blockchain: Trust economy", Deloitte University Press, 2017

## ***Blockchain – why use it?***

- Used to exchange value
- Can't be tampered with
- Crowdsource processing power
  - Suitable for scientific research
- Autonomous applications (like e.g. AirBnB)
- Tackle money laundering

